Indian Currency Classification

Leveraging Deep Learning Techniques for Automated Banknote Recognition

PORTFOLIO PROJECT

Data Science & Artificial Intelligence

Project Overview

Motivation, Problem Statement & Objectives

Introduction & Motivation

The Problem

Manual classification of currency is time-consuming and error-prone. With the 2016 demonetization and the introduction of the new Mahatma Gandhi series, automation is essential for banks and retail.

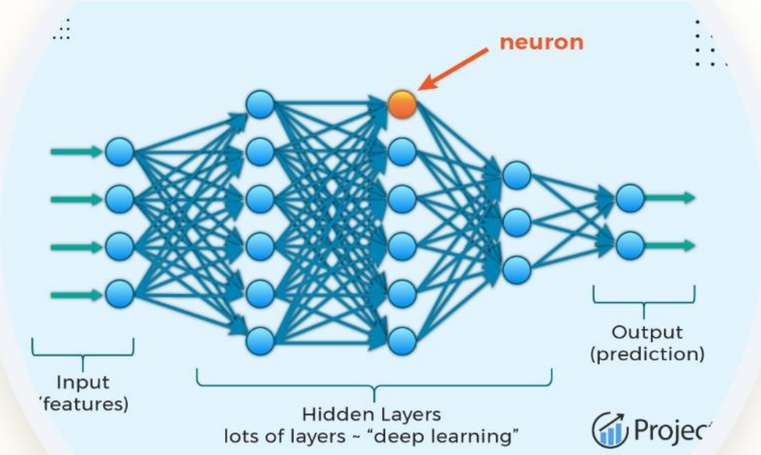
Key Objectives

Develop a robust deep learning model to accurately identify denominations (₹10 to ₹2000) under varying conditions like light, occlusion, and physical wear.

The Indian Banknote Dataset

- **Denominations:** 10, 20, 50, 100, 200, 500, 2000.
- **Volume:** 3,500+ images collected from multiple sources.
- **Variability:** Includes front/back views, angled shots, and folded notes.
- **Security:** Focus on identifying unique patterns and colors.

Proposed CNN Architecture



MobileNetV2 Transfer Learning

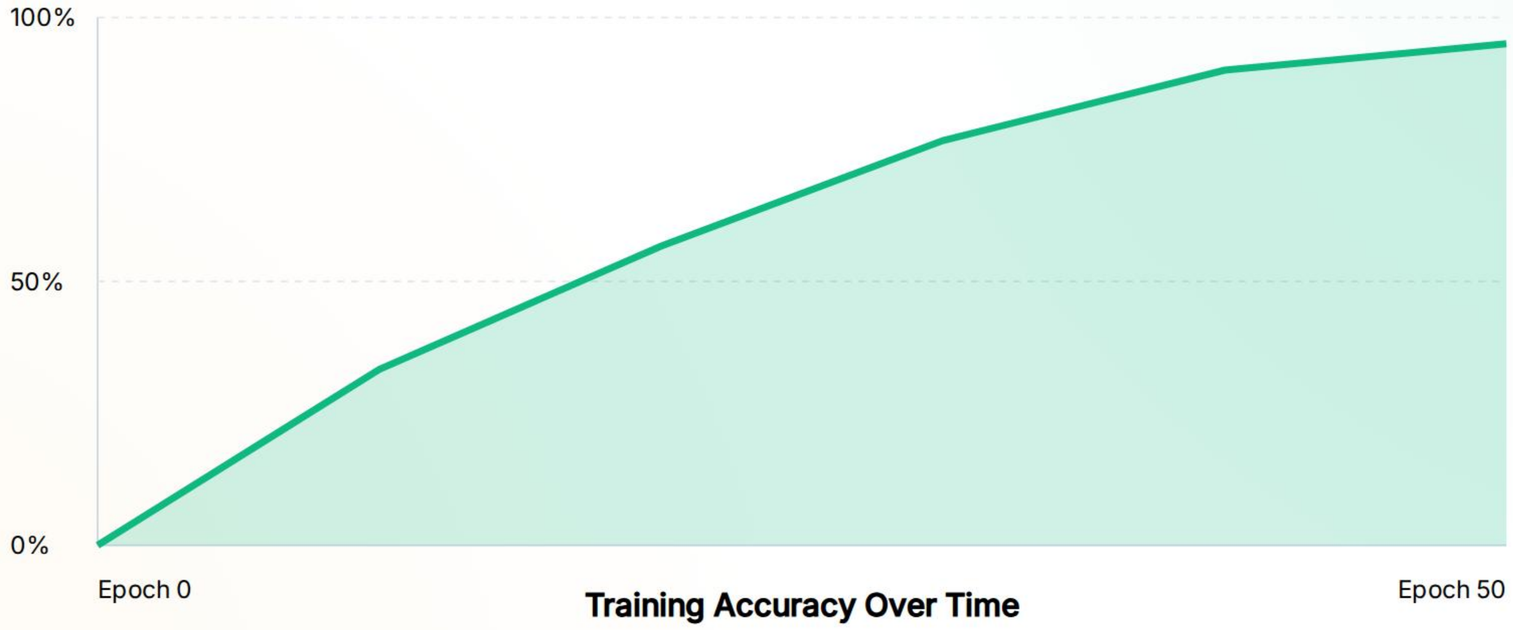
Chosen for its **lightweight** nature, making it ideal for mobile deployment. The model uses depth-wise separable convolutions to reduce parameters while maintaining high accuracy.

The pre-trained ImageNet weights were fine-tuned on our custom currency dataset to capture specific banknote features.

Image Preprocessing Pipeline

- **Standardization:** All images resized to 224×224 to fit the model input layer requirements.
- **Data Augmentation:** Applied random rotations, horizontal flips, and zoom to prevent overfitting and improve generalization.
- **Normalization:** Pixel values scaled to 01 for faster gradient descent convergence.
- **Noise Reduction:** Gaussian blurring used to minimize sensor noise from mobile captures.

Training Metrics & Convergence



Final validation accuracy reached 96.5% after 50 epochs with early stopping implemented.

Model Performance Comparison

MobileNetV2 outperformed other architectures in both speed and accuracy.

Denomination-wise Precision

Denomination	Precision	Recall	F1-Score
₹2000 Note	0.98	0.97	0.975
₹500 Note	0.95	0.96	0.955
₹200 Note	0.94	0.94	0.940
₹100 Note	0.96	0.95	0.955
₹50 / ₹20 / ₹10	0.93	0.92	0.925

Technology & Tools

Languages

Python 3.x for core development and scripting.

Frameworks

TensorFlow, Keras for DL model training.

Library

OpenCV for image processing & NumPy for math.

Application & Impact

Accessibility Solution

This technology can be integrated into mobile apps to assist the visually impaired in identifying currency notes through audio feedback.

Banking Automation

Enables smart ATMs and deposit machines to verify and sort currency without human intervention, reducing operational overhead.

Thank You!

Let's connect and discuss this project further.

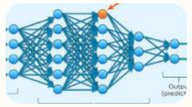
github.com/yourprofile

linkedin.com/in/yourname

Image Sources

https://st5.depositphotos.com/84897028/69601/i/450/depositphotos_696010198-stock-photo-100-500-rupee-notes-holding.jpg

Source: depositphotos.com



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Source: www.projectpro.io

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Source: news.microsoft.com